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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Redox Reactions

JEE Main Chemistry · Previous Year Questions with Answer Key

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Numerical 28 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

500 mL of $0.2 \mathrm{M} \mathrm{MnO}_4^-$ solution in basic medium when mixed with 500 mL of 1.5 M KI solution, oxidises iodide ions to liberate molecular iodine. This liberated iodine is then titrated with a standard $x \mathrm{M}$ thiosulphate solution in presence of starch till the end point. If 300 mL of thiosulphate was consumed, then the value of x is ____.

Open solution

Q2. JEE Main 2026 (Online) 28th January Morning Shift

500 mL of 1.2 M KI solution is mixed with 500 mL of $0.2 \mathrm{MKMnO}_4$ solution in basic medium. The liberated iodine was titrated with standard $0.1 \mathrm{M} \mathrm{Na}_2 \mathrm{S}_2 \mathrm{O}_3$ solution in the presence of starch indicator till the blue color disappeared. The volume (in L) of $\mathrm{Na}_2 \mathrm{S}_2 \mathrm{O}_3$ consumed is ____ . (Nearest integer)

Open solution

Q3. JEE Main 2026 (Online) 24th January Morning Shift

X and Y are the number of electrons involved, respectively during the oxidation of I^- to I_2 and S^{2-} to S by acidified $\mathrm{K}_2 \mathrm{Cr}_2 \mathrm{O}_7$. The value of $\mathrm{X}+\mathrm{Y}$ is ____ .

Open solution

Q4. JEE Main 2026 (Online) 23rd January Evening Shift

200 cc of $x \times 10^{-3} \mathrm{M}$ potassium dichromate is required to oxidise 750 cc of 0.6 M Mohr's salt solution in acidic medium.

Here $x=$ ____ .

Open solution

Q5. JEE Main 2025 (Online) 4th April Morning Shift

KMnO_4 acts as an oxidising agent in acidic medium. ' X ' is the difference between the oxidation states of Mn in reactant and product. ' Y ' is the number of ' d ' electrons present in the brown red precipitate formed at the end of the acetate ion test with neutral ferric chloride. The value of $\mathrm{X}+\mathrm{Y}$ is _____.

Open solution

Q6. JEE Main 2025 (Online) 22nd January Morning Shift

Some CO_2 gas was kept in a sealed container at a pressure of 1 atm and at 273 K . This entire amount of CO_2 gas was later passed through an aqueous solution of $\mathrm{Ca}(\mathrm{OH})_2$. The excess unreacted $\mathrm{Ca}(\mathrm{OH})_2$ was later neutralized with 0.1 M of 40 mL HCl . If the volume of the sealed container of CO_2 was x, then x is _____ cm^3 (nearest integer). [Given : The entire amount of $\mathrm{CO}_2(\mathrm{~g})$ reacted with exactly half the initial amount of $\mathrm{Ca}(\mathrm{OH})_2$ present in the aqueous solution.]

Open solution

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Q7. JEE Main 2024 (Online) 4th April Morning Shift

Only $2\,\mathrm{mL}$ of $\mathrm{KMnO_4}$ solution of unknown molarity is required to reach the end point of a titration of $20\,\mathrm{mL}$ of oxalic acid $(2\mathrm{M})$ in acidic medium. The molarity of $\mathrm{KMnO_4}$ solution should be _____ M.

Open solution

Q8. JEE Main 2024 (Online) 31st January Evening Shift

In the reaction of potassium dichromate, potassium chloride and sulfuric acid (conc.), the oxidation state of the chromium in the product is $(+)$ _____.

Open solution

Q9. JEE Main 2024 (Online) 31st January Evening Shift

Number of moles of $\mathrm{H^+}$ ions required by $1\,\mathrm{mole}$ of $\mathrm{MnO_4^-}$ to oxidise oxalate ion to $\mathrm{CO_2}$ is _____.

Open solution

Q10. JEE Main 2024 (Online) 30th January Evening Shift

Total number of species from the following which can undergo disproportionation reaction is _____.

$\mathrm{H_2O_2, ClO_3^-, P_4, Cl_2, Ag, Cu^{+1}, F_2, NO_2, K^+}$

Open solution

Q11. JEE Main 2024 (Online) 30th January Morning Shift

$\mathrm{2MnO_4^- + bI^- + cH_2O \rightarrow xI_2 + yMnO_2 + zOH^-}$

If the above equation is balanced with integer coefficients, the value of z is _____.

Open solution

Q12. JEE Main 2024 (Online) 29th January Evening Shift

The oxidation number of iron in the compound formed during brown ring test for $\mathrm{NO_3^-}$ ion is _____.

Open solution

Q13. JEE Main 2023 (Online) 13th April Evening Shift

See the following chemical reaction:

$\mathrm{Cr_2O_7^{2-} + XH^+ + 6\,Fe^{2+} \rightarrow YCr^{3+} + 6\,Fe^{3+} + ZH_2O}$

The sum of $\mathrm{X, Y}$ and Z is _____

Open solution

Q14. JEE Main 2023 (Online) 10th April Evening Shift

In alkaline medium, the reduction of permanganate anion involves a gain of _____ electrons.

Open solution

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Q15. JEE Main 2023 (Online) 8th April Evening Shift

The sum of oxidation state of the metals in $Fe(CO)_5$, VO^{2+} and WO_3 is _____.

Open solution

Q16. JEE Main 2023 (Online) 6th April Morning Shift

In ammonium - phosphomolybdate, the oxidation state of Mo is + _____

Open solution

Q17. JEE Main 2023 (Online) 1st February Morning Shift

Sum of oxidation states of bromine in bromic acid and perbromic acid is _____.

Open solution

Q18. JEE Main 2023 (Online) 31st January Morning Shift

The oxidation state of phosphorus in hypophosphoric acid is + _____.

Open solution

Q19. JEE Main 2022 (Online) 27th July Evening Shift

The normality of H_2SO_4 in the solution obtained on mixing $100\ mL$ of $0.1\ M\ H_2SO_4$ with $50\ mL$ of $0.1\ M\ NaOH$ is _____ $\times 10^{-1}$ N. (Nearest Integer)

Open solution

Q20. JEE Main 2022 (Online) 26th July Evening Shift

$20\ mL$ of $0.02\ M$ hypo solution is used for the titration of $10\ mL$ of copper sulphate solution, in the presence of excess of KI using starch as an indicator. The molarity of Cu^{2+} is found to be _____ $\times 10^{-2}$ M. [nearest integer]

Given : $2\,Cu^{2+} + 4\,I^- \rightarrow Cu_2I_2 + I_2$

$I_2 + 2\,S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$

Open solution

Q21. JEE Main 2022 (Online) 28th June Evening Shift

0.01 M $KMnO_4$ solution was added to 20.0 mL of 0.05 M Mohr's salt solution through a burette. The initial reading of 50 mL burette is zero. The volume of $KMnO_4$ solution left in the burette after the end point is _____ mL. (nearest integer)

Open solution

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Q22. JEE Main 2022 (Online) 25th June Evening Shift

The neutralization occurs when 10 mL of 0.1M acid 'A' is allowed to react with 30 mL of 0.05 M base $M(OH)_2$. The basicity of the acid 'A' is _____.

[M is a metal]

Open solution

Q23. JEE Main 2021 (Online) 27th July Evening Shift

10.0 mL of 0.05 M $KMnO_4$ solution was consumed in a titration with 10.0 mL of given oxalic acid dihydrate solution. The strength of given oxalic acid solution is _____ $\times 10^{-2}$ g/L.

(Round off to the nearest integer)

Open solution

Q24. JEE Main 2021 (Online) 26th February Morning Shift

Dichromate ion is treated with base, the oxidation number of Cr in the product formed is _____.

Open solution

Q25. JEE Main 2021 (Online) 25th February Morning Shift

0.4 g mixture of NaOH, Na_2CO_3 and some inert impurities was first titrated with $\frac{N}{10}$ HCl using phenolphthalein as an indicator, 17.5 mL of HCl was required at the end point. After this methyl orange was added and titrated. 1.5 mL of same HCl was required for the next end point. The weight percentage of Na_2CO_3 in the mixture is _____. (Rounded off to the nearest integer)

Open solution

Q26. JEE Main 2020 (Online) 5th September Evening Slot

The volume, in mL, of 0.02 M $K_2Cr_2O_7$ solution required to react with 0.288 g of ferrous oxalate in acidic medium is _____.
(Molar mass of Fe = 56 g mol^{-1})

Open solution

Q27. JEE Main 2020 (Online) 4th September Evening Slot

A 100 mL solution was made by adding 1.43 g of $Na_2CO_3.xH_2O$. The normality of the solution is 0.1 N. The value of x is _____.

(The atomic mass of Na is 23 g/mol)

Open solution

Q28. JEE Main 2020 (Online) 2nd September Evening Slot

The oxidation states of transition metal atoms in $K_2Cr_2O_7$, $KMnO_4$ and K_2FeO_4, respectively, are x, y and z. The sum of x, y and z is _____.

Open solution

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MCQ (Single Correct Answer) 52 questions

Q29. JEE Main 2026 (Online) 4th April Morning Shift

In order to oxidise a mixture of 1 mole each of $FeC_2O_4, Fe_2(C_2O_4)_3, FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium, the number of moles of $KMnO_4$ required is

(A) 3 (B) 2 (C) 5 (D) 7

Open solution

Q30. JEE Main 2026 (Online) 2nd April Morning Shift

Given below are two statements:

Statement (I): Oxidising power of halogens decreases in the order $F_2 > Cl_2 > Br_2 > I_2$, which is the basis of "Layer test".

Statement (II): "Layer test" to identify Br_2 and I_2 in aqueous solution involves the oxidation of bromide or iodide into Br_2 or I_2 respectively with Cl_2, which is a type of displacement redox reaction.

In the light of the above statements, choose the **correct answer** from the options given below:

(A) Both **Statement I** and **Statement II** are true

(B) Both **Statement I** and **Statement II** are false

(C) **Statement I** is true but **Statement II** is false

(D) **Statement I** is false but **Statement II** is true

Open solution

Q31. JEE Main 2026 (Online) 24th January Evening Shift

One mole of $Cl_2(g)$ was passed into 2 L of cold 2 M KOH solution. After the reaction, the concentrations of Cl^-, ClO^- and OH^- are respectively (assume volume remains constant)

(A) $0.5M, 0.5M, 0.5M$ (B) $1M, 1M, 1M$

(C) $0.5M, 0.5M, 1M$ (D) $0.75M, 0.75M, 1M$

Open solution

Q32. JEE Main 2026 (Online) 23rd January Evening Shift

The oxidation state of chromium in the final product formed in the reaction between KI and acidified $K_2Cr_2O_7$ solution is :

(A) +2 (B) +3 (C) +4 (D) +6

Open solution

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Q33. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solutions. According to equation

$5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

Identify the correct statements :

(A) 200 mL of KI solution reacts with 0.004 mol of KIO_3

(B) 200 mL of KI solution reacts with 0.006 mol of H_2SO_4

(C) 0.5 L of KI solution produced 0.005 mol of I_2

(D) Equivalent weight of KIO_3 is equal to ($\frac{\text{Molecular weight}}{5}$)

Choose the **correct** answer from the options given below :

(A) (C) and (D) only **(B)** (A) and (B) only **(C)** (B) and (C) only **(D)** (A) and (D) only

Q34. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Given below are two statements:

Statement I: In the oxalic acid vs $KMnO_4$ (in the presence of dil H_2SO_4) titration the solution needs to be heated initially to $60^\circ C$, but no heating is required in Ferrous ammonium sulphate (FAS) vs $KMnO_4$ titration (in the presence of dil H_2SO_4)

Statement II: In oxalic acid vs $KMnO_4$ titration, the initial formation of $MnSO_4$ takes place at high temperature, which then acts as catalyst for further reaction. In the case of FAS vs $KMnO_4$, heating oxidizes Fe^{2+} into Fe^{3+} by oxygen of air and error may be introduced in the experiment.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

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Q35. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Match the List - I with List - II

	List - I (Redox Reaction)		List - II (Type of Redox Reaction)
(A)	$CH_{4(g)} + 2O_{2(g)} \xrightarrow{\Delta} CO_{2(g)} + 2H_2O_{(l)}$	(I)	Disproportionation reaction
(B)	$2NaH_{(s)} \xrightarrow{\Delta} 2Na_{(s)} + H_{2(g)}$	(II)	Combination reaction
(C)	$V_2O_{5(s)} + 5Ca_{(s)} \xrightarrow{\Delta} 2\,V_{(s)} + 5CaO_{(s)}$	(III)	Decomposition reaction
(D)	$2H_2O_{2(aq)} \xrightarrow{\Delta} 2H_2O_{(l)} + O_{2(g)}$	(IV)	Displacement reaction

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II

(B) A-IV, B-I, C-II, D-III

(C) A-II, B-III, C-I, D-IV

(D) A-II, B-III, C-IV, D-I

Q36. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Which of the following oxidation reactions are carried out by both $K_2Cr_2O_7$ and $KMnO_4$ in acidic medium?

A. $I^- \rightarrow I_2$

B. $S^{2-} \rightarrow S$

C. $Fe^{2+} \rightarrow Fe^{3+}$

D. $I^- \rightarrow IO_3^-$

E. ${S_2O_3}^{2-} \rightarrow SO_4^{2-}$

Choose the correct answer from the options given below :

(A) A, B and C Only **(B)** C, D and E Only **(C)** B, C and D Only **(D)** A, D and E Only

Q37. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

The species which does not undergo disproportionation reaction is :

(A) ClO_4^- **(B)** ClO_3^- **(C)** ClO^- **(D)** ClO_2^-

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Q38. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Thiosulphate reacts differently with iodine and bromine in the reactions given below:

$2\,S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$

$S_2O_3^{2-} + 5Br_2 + 5H_2O \rightarrow 2SO_4^{2-} + 4Br^- + 10H^+$

Which of the following statements justifies the above dual behaviour of thiosulphate?

(A) Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reactions

(B) Bromine is a weaker oxidant than iodine

(C) Bromine is a stronger oxidant than iodine

(D) Bromine undergoes oxidation and iodine undergoes reduction in these reactions

Q39. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

The number of ions from the following that are expected to behave as oxidising agent is :

$Sn^{4+}, Sn^{2+}, Pb^{2+}, Tl^{3+}, Pb^{4+}, Tl^+$

(A) 3 (B) 4 (C) 2 (D) 1

Q40. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

Match List I with List II.

	LIST I Reaction		**LIST II Type of redox reaction**
A.	$N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$	I.	Decomposition
B.	$2\,Pb(NO_3)_{2(s)} \rightarrow 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$	II.	Displacement
C.	$2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq.)} + H_{2(g)}$	III.	Disproportionation
D.	$2NO_{2(g)} + 2OH^-_{(aq.)} \rightarrow NO^-_{2(aq.)} + NO^-_{3(aq.)} + H_2O_{(l)}$	IV.	Combination

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(B) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(C) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(D) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Q41. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

When MnO_2 and H_2SO_4 are added to a salt (A), a greenish yellow gas is liberated. The salt (A) is :

(A) NH_4Cl (B) CaI_2 (C) KNO_3 (D) $NaBr$

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Q42. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

Which of the following reactions are disproportionation reactions?

(A) $Cu^{+} \rightarrow Cu^{2+} + Cu$

(B) $3MnO_4^{2-} + 4H^{+} \longrightarrow 2MnO_4^{-} + MnO_2 + 2H_2O$

(C) $2KMnO_4 \longrightarrow K_2MnO_4 + MnO_2 + O_2$

(D) $2MnO_4^{-} + 3Mn^{2+} + 2H_2O \longrightarrow 5MnO_2 + 4H^{+}$

Choose the correct answer from the options given below :

(A) (A), (B) **(B)** (A), (D) **(C)** (B), (C), (D) **(D)** (A), (B), (C)

Q43. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

In acidic medium, $K_2Cr_2O_7$ shows oxidising action as represented in the half reaction:

$$Cr_2O_7^{2-} + XH^{+} + Ye^{\ominus} \rightarrow 2\,A + ZH_2O$$

X, Y, Z and A are respectively :

(A) $14, 7, 6$ and Cr^{3+} **(B)** $14, 6, 7$ and Cr^{3+}

(C) $8, 4, 6$ and Cr_2O_3 **(D)** $8, 6, 4$ and Cr_2O_3

Q44. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Chlorine undergoes disproportionation in alkaline medium as shown below :

$aCl_{2(g)} + bOH^{-}_{(aq)} \rightarrow cClO^{-}_{(aq)} + dCl^{-}_{(aq)} + eH_2O_{(l)}$

The values of a, b, c and d in a balanced redox reaction are respectively :

(A) 3, 4, 4 and 2 **(B)** 1, 2, 1 and 1 **(C)** 2, 4, 1 and 3 **(D)** 2, 2, 1 and 3

Q45. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

In alkaline medium, MnO_4^{-} oxidises I^{-} to

(A) I_2 **(B)** IO_3^{-} **(C)** IO^{-} **(D)** IO_4^{-}

Q46. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

Which of the following cannot function as an oxidising agent?

(A) SO_4^{2-} **(B)** MnO_4^{-} **(C)** N^{3-} **(D)** BrO_3^{-}

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Q47. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

Given below are two statements:

Statement I : In redox titration, the indicators used are sensitive to change in pH of the solution.

Statement II : In acid-base titration, the indicators used are sensitive to change in oxidation potential.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Both Statement I and Statement II are incorrect

(B) Statement I is correct but Statement II is incorrect

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are correct

Q48. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

$2\mathrm{IO}_3^- + x\mathrm{I}^- + 12\mathrm{H}^+ \rightarrow 6\mathrm{I}_2 + 6\mathrm{H}_2\mathrm{O}$

What is the value of x ?

(A) 10 (B) 2 (C) 12 (D) 6

Q49. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

Which of the following options are correct for the reaction

$2[\mathrm{Au(CN)_2}]^-(\mathrm{aq}) + \mathrm{Zn(s)} \rightarrow 2\mathrm{Au(s)} + [\mathrm{Zn(CN)_4}]^{2-}(\mathrm{aq})$

A. Redox reaction

B. Displacement reaction

C. Decomposition reaction

D. Combination reaction

Choose the correct answer from the options given below:

(A) A and B only (B) C and D only (C) A only (D) A and D only

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Q50. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$.

Assertion A: Phenolphthalein is a pH dependent indicator, remains colourless in acidic solution and gives pink colour in basic medium.

Reason R: Phenolphthalein is a weak acid. It doesn't dissociate in basic medium.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$.

(C) A is true but R is false.

(D) A is false but R is true.

Q51. JEE Main 2022 (Online) 26th July Morning Shift

Open solution

Which of the given reactions is not an example of disproportionation reaction?

(A) $2\mathrm{H_2O_2} \rightarrow 2\mathrm{H_2O} + \mathrm{O_2}$

(B) $2\mathrm{NO_2} + \mathrm{H_2O} \rightarrow \mathrm{HNO_3} + \mathrm{HNO_2}$

(C) $\mathrm{MnO_4^-} + 4\mathrm{H^+} + 3\mathrm{e^-} \rightarrow \mathrm{MnO_2} + 2\mathrm{H_2O}$

(D) $3\mathrm{MnO_4^{2-}} + 4\mathrm{H^+} \rightarrow 2\mathrm{MnO_4^-} + \mathrm{MnO_2} + 2\mathrm{H_2O}$

Q52. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

Which one of the following is an example of disproportionation reaction ?

(A) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

(B) $MnO_4^- + 4H^+ + 4e^- \rightarrow MnO_2 + 2H_2O$

(C) $10I^- + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5I_2$

(D) $8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

Q53. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

The oxidation states of 'P' in $H_4P_2O_7$, $H_4P_2O_5$ and $H_4P_2O_6$, respectively, are :

(A) 7, 5 and 6 (B) 5, 4 and 3 (C) 5, 3 and 4 (D) 6, 4 and 5

Q54. JEE Main 2021 (Online) 25th July Evening Shift

Open solution

Identify the process in which change in the oxidation state is five :

(A) $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$ (B) $MnO_4^- \rightarrow Mn^{2+}$

(C) $CrO_4^{2-} \rightarrow Cr^{3+}$ (D) $C_2O_4^{2-} \rightarrow 2CO_2$

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Q55. JEE Main 2021 (Online) 18th March Evening Shift

The oxidation states of nitrogen in NO, NO_2, N_2O and NO_3^- are in the order of :

(A) $N_2O > NO_2 > NO > NO_3^-$

(B) $NO > NO_2 > N_2O > NO_3^-$

(C) $NO_2 > NO_3^- > NO > N_2O$

(D) $NO_3^- > NO_2 > NO > N_2O$

Open solution

Q56. JEE Main 2021 (Online) 16th March Evening Shift

Statement I : Sodium hydride can be used as an oxidising agent.

Statement II : The lone pair of electrons on nitrogen in pyridine makes it basic.

Choose the CORRECT answer from the options given below :

(A) Statement I is false but statement II is true

(B) Both statement I and statement II are true

(C) Statement I is true but statement II is false

(D) Both statement I and statement II are false

Open solution

Q57. JEE Main 2020 (Online) 2nd September Morning Slot

While titrating dilute HCl solution with aqueous NaOH, which of the following will not be required?

(A) Pipette and distilled water

(B) Clamp and phenolphthalein

(C) Burette and porcelain tile

(D) Bunsen burner and measuring cylinder

Open solution

Q58. JEE Main 2020 (Online) 9th January Morning Slot

The compound that cannot act both as oxidising and reducing agent is :

(A) H_3PO_4

(B) H_2SO_3

(C) H_2O_2

(D) HNO_2

Open solution

Q59. JEE Main 2020 (Online) 7th January Evening Slot

The redox reaction among the following is :

(A) reaction of H_2SO_4 with NaOH.

(B) formation of ozone from atmospheric oxygen in the presence of sunlight.

(C) combination of dinitrogen with dioxygen at 2000 K

(D) reaction of $[Co(H_2O)_6]Cl_3$ With $AgNO_3$

Open solution

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Q60. JEE Main 2020 (Online) 7th January Morning Slot

Oxidation number of potassium in K_2O, K_2O_2 and KO_2 respectively is :

(A) +1, +2 and + 4

(B) +1, +1 and + 1

(C) +2, +1 and $+\frac{1}{2}$

(D) +1, +4, and +2

Open solution

Q61. JEE Main 2019 (Online) 12th April Morning Slot

An example of a disproportionation reaction is :

(A) $2NaBr + Cl_2 \rightarrow 2NaCl + Br_2$

(B) $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$

(C) $2CuBr \rightarrow CuBr_2 + Cu$

(D) $2MnO_4^- + 10I^- + 16H^+ \rightarrow 2Mn^{2+} + 5I_2 + 8H_2O$

Open solution

Q62. JEE Main 2019 (Online) 9th April Morning Slot

The correct order of the oxidation states of nitrogen in NO, N_2O, NO_2 and N_2O_3 is :

(A) $NO_2 < NO < N_2O_3 < N_2O$

(B) $NO_2 < N_2O_3 < NO < N_2O$

(C) $N_2O < NO < N_2O_3 < NO_2$

(D) $N_2O < N_2O_3 < NO < NO_2$

Open solution

Q63. JEE Main 2019 (Online) 8th April Morning Slot

In order to oxidise a mixture of one mole of each of FeC_2O_4, $Fe_2(C_2O_4)_3$, $FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium, the number of moles of $KMnO_4$ required is :

(A) 1.5 (B) 3 (C) 2 (D) 1

Open solution

Q64. JEE Main 2019 (Online) 12th January Morning Slot

50 mL of 0.5 M oxalic acid is needed to neutralize 25 mL of sodium hydroxide solution. The amount of NaOH in 50 mL of the given sodium hydroxide solution is -

(A) 20 g (B) 4 g (C) 80 g (D) 10 g

Open solution

Q65. JEE Main 2019 (Online) 12th January Morning Slot

The hardness of a water sample (in terms of equivalents of $CaCO_3$) containing 10^{-3} M $CaSO_4$ is

(Molar mass of $CaSO_4$ = 136 g mol^{-1})

(A) 90 ppm (B) 100 ppm (C) 50 ppm (D) 10 ppm

Open solution

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Q66. JEE Main 2019 (Online) 10th January Evening Slot

In the reaction of oxalate with permanganate in acidic medium, the number of electrons involved in producing one molecule of CO_2 is :

(A) 10 (B) 2 (C) 1 (D) 5

Open solution

Q67. JEE Main 2017 (Offline)

Which of the following reactions is an example of a redox reaction?

(A) $XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$ (B) $XeF_6 + 2H_2O \rightarrow XeO_2F_2 + 4HF$

(C) $XeF_4 + O_2F_2 \rightarrow XeF_6 + O_2$ (D) $XeF_2 + PF_5 \rightarrow [XeF]^+PF_6^-$

Open solution

Q68. JEE Main 2016 (Online) 10th April Morning Slot

The volume of 0.1N dibasic acid sufficient to neutralize 1 g of a base that furnishes 0.04 mole of OH^- in aqueous solution is :

(A) 200 mL (B) 400 mL (C) 600 mL (D) 800 mL

Open solution

Q69. JEE Main 2015 (Online) 11th April Morning Slot

Choose the incorrect formula out of the four compounds for an element X below :

(A) X_2Cl_3 (B) X_2O_3 (C) $X_2(SO_4)_3$ (D) XPO_4

Open solution

Q70. JEE Main 2014 (Online) 19th April Morning Slot

Consider the reaction :

$$H_2SO_{3(aq)} + Sn^{4+}_{(aq)} + H_2O_{(l)} \rightarrow Sn^{2+}_{(aq)} + HSO^-_{4(aq)} + 3H^+_{(aq)}$$

Which of the following statements is correct ?

(A) Sn^{4+} is the oxidizing agent because it undergoes oxidation

(B) Sn^{4+} is the reducing agent because it undergoes oxidation

(C) H_2SO_3 is the reducing agent because it undergoes oxidation

(D) H_2SO_3 is the reducing agent because it undergoes reduction

Open solution

Q71. JEE Main 2014 (Online) 19th April Morning Slot

How many electrons are involved in the following redox reaction ?

$$Cr_2O_7^{2-} + Fe^{2+} + C_2O_4^{2-} \rightarrow Cr^{3+} + Fe^{3+} + CO_2 \text{ (Unbalanced)}$$

(A) 3 (B) 4 (C) 6 (D) 5

Open solution

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Q72. JEE Main 2014 (Online) 19th April Morning Slot

Amongst the following, identify the species with an atom in +6 oxidation state :

(A) $[MnO_4]^-$ (B) $[Cr(CN)_6]^{3-}$ (C) Cr_2O_3 (D) CrO_2Cl_2

Open solution

Q73. JEE Main 2014 (Offline)

In which of the following reactions H_2O_2 acts as a reducing agent?

1. $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$
2. $H_2O_2 - 2e^- \rightarrow O_2 + 2H^+$
3. $H_2O_2 + 2e^- \rightarrow 2OH^-$
4. $H_2O_2 + 2OH^- - 2e^- \rightarrow O_2 + 2H_2O$

(A) 1, 2 (B) 3, 4 (C) 1, 3 (D) 2, 4

Open solution

Q74. JEE Main 2013 (Online) 25th April Morning Slot

Which one of the following cannot function as an oxidising agent?

(A) I^- (B) S(s) (C) NO_3^- (aq) (D) $Cr_2O_7^{2-}$

Open solution

Q75. JEE Main 2013 (Online) 22th April Morning Slot

Oxidation state of sulphur in anions SO_3^{2-}, $S_2O_4^{2-}$ and $S_2O_6^{2-}$ increases in the orders :

(A) $S_2O_6^{2-} < S_2O_4^{2-} < SO_3^{2-}$ (B) $SO_3^{2-} < S_2O_4^{2-} < S_2O_6^{2-}$

(C) $S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}$ (D) $S_2O_4^{2-} < S_2O_6^{2-} < SO_3^{2-}$

Open solution

Q76. JEE Main 2013 (Online) 9th April Morning Slot

Given : $XNa_2HAsO_3 + YNaBrO_3 + ZHCl \rightarrow NaBr + H_3AsO_4 + NaCl$

The values of X, Y and Z in the above redox reaction are respectively :

(A) 2, 1, 2 (B) 2, 1, 3 (C) 3, 1, 6 (D) 3, 1, 4

Open solution

Q77. JEE Main 2013 (Offline)

Consider the following reaction:

$xMnO_4^- + yC_2O_4^{2-} + zH^+ \rightarrow xMn^{2+} + 2yCO_2 + \frac{z}{2}H_2O$

The values of x, y and z in the reaction are, respectively :

(A) 5, 2 and 16 (B) 2, 5 and 8 (C) 2, 5 and 16 (D) 5, 2 and 8

Open solution

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Q78. AIEEE 2012 (Online) 12th May

In the following balanced reaction,

$$X\,\mathrm{MnO_4^-} + Y\,\mathrm{C_2O_4^{2-}} + Z\,\mathrm{H^+} \rightleftharpoons X\,\mathrm{Mn^{2+}} + 2Y\,\mathrm{CO_2} + \frac{Z}{2}\,\mathrm{H_2O}$$

values of X, Y and Z respectively are

(A) 2, 5, 16 (B) 8, 2, 5 (C) 5, 2, 16 (D) 5, 8, 4

Open solution

Q79. AIEEE 2006

Which of the following chemical reactions depicts the oxidizing behaviour of H_2SO_4?

(A) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$

(B) $2PCl_5 + H_2SO_4 \rightarrow 2POCl_3 + 2HCl + SO_2Cl_2$

(C) $2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$

(D) $Ca(OH)_2 + H_2SO_4 \rightarrow CaSO_4 + 2H_2O$

Open solution

Q80. AIEEE 2002

Which of the following is a redox reaction ?

(A) $NaCl + KNO_3 \rightarrow NaNO_3 + KCl$

(B) $CaC_2O_4 + 2HCl \rightarrow CaCl_2 + H_2C_2O_4$

(C) $Mg(OH)_2 + 2NH_4Cl \rightarrow MgCl_2 + 2NH_4OH$

(D) $Zn + 2AgCN \rightarrow 2Ag + Zn(CN)_2$

Open solution

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Answer Key

Numerical

1. **1**	2. **3**	3. **12**	4. **375**	5. **10**
6. **45**	7. **8**	8. **6**	9. **8**	10. **6**
11. **8**	12. **1**	13. **23**	14. **3**	15. **10**
16. **6**	17. **12**	18. **4**	19. **1**	20. **4**
21. **30**	22. **3**	23. **1575**	24. **6**	25. **4**
26. **50**	27. **10**	28. **19**		

MCQ (Single Correct Answer)

29. **B**	30. **A**	31. **C**	32. **B**	33. **D**
34. **C**	35. **D**	36. **A**	37. **A**	38. **C**
39. **C**	40. **A**	41. **A**	42. **A**	43. **B**
44. **B**	45. **B**	46. **C**	47. **A**	48. **A**
49. **A**	50. **C**	51. **C**	52. **A**	53. **C**
54. **B**	55. **D**	56. **A**	57. **D**	58. **A**
59. **C**	60. **B**	61. **C**	62. **C**	63. **C**
64. **B**	65. **B**	66. **C**	67. **C**	68. **B**
69. **A**	70. **C**	71. **C**	72. **D**	73. **D**
74. **A**	75. **C**	76. **C**	77. **C**	78. **A**
79. **C**	80. **D**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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